WARING DECOMPOSITION OF DERIVATIVES AND ITS APPLICATIONS

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Abstract. Computing high-order derivatives with neural networks is often computationally and memory intensive, primarily due to the huge computational graphs caused by automatic differentiation. In this work, we decompose the derivative into directional Taylor-mode automatic differentiation. We establish a fundamental connection between derivative extraction and polynomial algebra, proving that the minimal number of directional evaluations required to recover a target partial derivative equals the Waring rank of its corresponding monomial. We present a roots-of-unity formula that strictly attains this theoretical lower bound. Experiments show that our method significantly outperforms traditional automatic differentiation in speed and memory efficiency.

Key words. high-order derivatives, Taylor-mode automatic differentiation, Waring decomposition, apolarity

AMS subject classifications. 65D25, 14N07

1. Introduction. In this work, we focus on computing monomial differential operators for analytic neural networks. Suppose $u\colon \mathbb{C}^d \to \mathbb{C}$ is analytic. Given a multi-index $\nu = (\nu_1, \ldots, \nu_d)$ of order $p = |\nu|$, the corresponding monomial differential operator is

$$\partial^\nu u(\boldsymbol{x}) := \partial_1^{\nu_1} \cdots \partial_d^{\nu_d} u(\boldsymbol{x}). \tag{1}$$

The standard evaluation of (1) is automatic differentiation [1, 9, 22]. However, when the order p or the dimension d is high, this evaluation is expensive in both memory and computation: the derivative tensor has size $O(d^p)$ and the computation graph has size $O(2^{p-1}L)$, where L is the number of operations in the forward computation graph [26]. This bottleneck has motivated other exact evaluations. Taylor-mode automatic differentiation computes directional derivatives of order p in one pass [2, 9]. Univariate Taylor series recover the full derivative tensor of that order [8]. In addition, a structured recurrence evaluates one differential operator [20]. These exact alternatives either reconstruct the full derivative tensor or evaluate only a directional coefficient or one structured operator, rather than a shortest combination for one monomial operator.

Several efforts have been made to avoid this computational cost. One line of work uses randomization to amortize the evaluation over the optimization process. Stochastic dimension gradient descent computes partial derivatives only for a minibatch of sampled coordinates [13]. Hutchinson's estimator [14], applied through Hessian-vector products [23], estimates traces of high-order operators [11]. Others bypass automatic differentiation completely. Finite differences replace each extra derivative by nearby evaluations of the same network [4, 6]. Randomized smoothing expresses derivatives as an expectation over a Gaussian random variable via Stein's identity [10, 28]. The stochastic Taylor derivative estimator chooses the input tangents of Taylor-mode automatic differentiation so as to estimate an arbitrary contraction of the derivative tensor [26]. However, compared with exact automatic differentiation, these estimators are sensitive to their hyperparameters, and the variance of the estimate is of great concern [6, 10, 12, 14, 26].

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In this paper, we use the Waring decomposition of monomials [3, 19] to obtain a Waring decomposition of derivatives, and from this we give an efficient exact method for computing derivatives. We give a decomposition of (1) whose number of terms attains the minimum,

$$\partial^\nu u(\boldsymbol{x}) = \sum_{r=1}^{R} c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r). \tag{2}$$

We use this decomposition to evaluate the mixed partial exactly. Each term has the form of a directional Taylor coefficient

$$T_p(\boldsymbol{x}; \boldsymbol{v}) = \frac{1}{p!}\, \frac{d^p}{d\tau^p}\, u(\boldsymbol{x} + \tau \boldsymbol{v})\Big|_{\tau=0}. \tag{3}$$

Specifically, our main contributions are as follows.

1. We prove that a decomposition of the form (2) is a Waring decomposition of the corresponding derivative: it is equivalent to a Waring decomposition of the monomial, and the least number of terms is the Waring rank of that monomial [3, 19].
2. We give an algorithm that attains this number of terms by a roots-of-unity formula, and thereby an efficient exact method for (1).
3. We apply the method to manufactured collocation residuals. The resulting evaluation uses less memory and less time than backpropagation applied p times.

The remainder of the paper is organized as follows. Section 2 states the evaluation problem and recalls Taylor-mode automatic differentiation. Section 3 proves the equivalence with monomial Waring rank and records the algorithm. Section 4 compares the implementation with backpropagation applied p times on manufactured collocation residuals [17, 18, 24, 25, 27]. Section 5 concludes. The roots-of-unity formula, the schedules used in the experiments, and the argument for the rank bound are collected in Appendix A.

2. Preliminaries.

2.1. The evaluation problem. Let $u_\theta \colon \mathbb{R}^d \to \mathbb{C}$ be smooth; real-valued functions are included as a special case. Let $\nu = (\nu_1, \ldots, \nu_d) \in \mathbb{Z}^d_{\geq 0}$ be a multi-index of order $p = |\nu|$, and write ∂^ν as in (1). We evaluate $\partial^\nu u_\theta$ at a point, exactly, and we differentiate the value with respect to θ. The same derivative is required at many points on every step, so the cost that matters is one evaluation, amortized over a batch.

A differential operator of order p is a linear combination of monomials ∂^α. Section 4 times the method on repeated evaluations of such an operator. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with boundary $\Gamma := \partial\Omega$, and consider

$$\mathcal{P}u = f \quad \text{in } \Omega, \qquad \mathcal{B}u = g \quad \text{on } \Gamma, \tag{4}$$

with $\mathcal{P}$ of order p and $\mathcal{B}$ the boundary conditions. Given interior points $(\boldsymbol{x}_{r,i})_{i=1}^{n_r} \subset \Omega$ and boundary points $(\boldsymbol{x}_{b,i})_{i=1}^{n_b} \subset \Gamma$, a collocation fit [17, 24] of (4) minimizes

$$\mathcal{L}(\theta) := \frac{\lambda_r}{n_r} \sum_{i=1}^{n_r} \left|\mathcal{P}u_\theta(\boldsymbol{x}_{r,i}) - f(\boldsymbol{x}_{r,i})\right|^2 + \frac{\lambda_b}{n_b} \sum_{i=1}^{n_b} \left|\mathcal{B}u_\theta(\boldsymbol{x}_{b,i}) - g(\boldsymbol{x}_{b,i})\right|^2. \tag{5}$$

Expanding $\mathcal{P}$ into monomials turns one residual evaluation in (5) into several derivatives $\partial^\alpha u_\theta$ of the same function on the same batch. The operator Δ^m expands into $\binom{d+m-1}{m}$ distinct monomials: the pure terms $\partial_{x_i}^{2m}$ and, for $m \geq 2$, mixed terms such as $\partial_{x_i}^{2k}\partial_{x_j}^{2m-2k}$.

2.2. Taylor-mode evaluation of directional derivatives. Let $u\colon \mathbb{R}^d \to \mathbb{C}$ be sufficiently smooth near a fixed point $\boldsymbol{x}$. For a direction $\boldsymbol{v} \in \mathbb{R}^d$, the directional Taylor coefficient of order p at $\boldsymbol{x}$ along $\boldsymbol{v}$, already written as (3), is

$$T_p(\boldsymbol{x};\boldsymbol{v}) := \frac{1}{p!}\,\frac{d^p}{d\tau^p}\,u(\boldsymbol{x}+\tau\boldsymbol{v})\Big|_{\tau=0} = \frac{1}{p!}\,D^p u(\boldsymbol{x})[\boldsymbol{v},\ldots,\boldsymbol{v}]. \tag{6}$$

Taylor-mode automatic differentiation evaluates (6) in one forward pass [2, 9]. Each intermediate activation $y \in \mathbb{C}^B$ over a batch of B points is replaced by the length-$(p+1)$ tuple $(y_0,\ldots,y_p)$ of its truncated Taylor coefficients

$$y_k = \frac{1}{k!}\,\frac{d^k y}{d\tau^k}\Big|_{\tau=0}. \tag{7}$$

For a multilayer perceptron $u = \phi_L \circ A_L \circ \cdots \circ \phi_1 \circ A_1$ with affine maps A_ℓ and entrywise activations ϕ_ℓ, an affine layer $y = Wx + b$ propagates each order independently,

$$y_0 = Wx_0 + b, \quad y_k = Wx_k \quad (k \geq 1), \tag{8}$$

and each nonlinearity is propagated together with one auxiliary jet that closes its derivative recurrence. For tanh the auxiliary jet is $z = 1 - y^2$, initialized by $y_0 = \tanh(x_0)$ and $z_0 = 1 - y_0^2$, and

$$y_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,z_j\,x_{k-j}, \quad z_k = -\sum_{j=0}^{k} y_j\,y_{k-j}, \quad k \geq 1. \tag{9}$$

For sinh the auxiliary jet is $z = \cosh(x)$, initialized by $y_0 = \sinh(x_0)$ and $z_0 = \cosh(x_0)$, and

$$y_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,z_j\,x_{k-j}, \quad z_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,y_j\,x_{k-j}, \quad k \geq 1. \tag{10}$$

The recurrence (10) has the same coupled form as the one for sin and cos, except that the cosh update carries no sign flip. The implementation also contains the recurrences for sin and for the logistic sigmoid, which the verification tests of Section 4 use. Every coefficient y_k is an ordinary tensor, and the propagation stores $p+1$ coefficient tensors per activation without nesting input-gradient graphs.

3. Rank-optimal directional schedules. The argument of this section is short. Its four steps are as follows. First, the directional Taylor coefficients of order p, read as a function of the direction rather than of the point, are the coefficients of a homogeneous polynomial of degree p, and the target derivative is one of them. Second, a formula that reconstructs the derivative from finitely many directional evaluations is therefore a linear identity among such coefficients, which reduces the design problem to a finite system of moment conditions. Third, pairing each direction with the polynomial variable turns that system into a single polynomial identity: the monomial belonging to the target derivative, written as a linear combination of pth powers of linear forms. Fourth, the least number of terms in such a combination is by definition the Waring rank of that monomial, a quantity known in closed form, and one set of minimizing linear forms consists of roots of unity. The four steps together replace a search for a reconstruction formula by a rank computation, and because the outcome is a minimum, it certifies that no shorter linear combination of those coefficients exists.

We introduce some notation used throughout this section. Let $\mathbb{N}_0 := \{0, 1, 2, \ldots\}$, and let $\mathbb{C}^d$ be the d-dimensional complex space. The *active coordinates* of ν are the indices k with $\nu_k > 0$; we relabel them as $i_0, \ldots, i_n$ and set $a_j := \nu_{i_j}$, ordered so that $1 \le a_0 \le \cdots \le a_n$, with ties broken by the original coordinate index. The ordering singles out a coordinate i_0 of least exponent, which is the coordinate the rank formula treats separately. We write $a := (a_0, \ldots, a_n)$, and we record for later use that

$$p = \sum_{j=0}^{n} a_j, \quad \nu! = \prod_{k=1}^{d} \nu_k! = \prod_{j=0}^{n} a_j!, \quad m_j := a_j + 1. \tag{11}$$

The *active subspace* is $V := \operatorname{span}\{e_{i_0}, \ldots, e_{i_n}\} \subset \mathbb{R}^d$, and $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$ is its complexification. Since ∂^ν is insensitive to coordinates outside the active set, all directional probes below may be taken in $V_{\mathbb{C}}$. The space of homogeneous polynomials of degree p in $\boldsymbol{z} = (z_0, \ldots, z_n)$ is denoted $\mathbb{C}[\boldsymbol{z}]_p$, and $[\boldsymbol{z}^a]F$ denotes the coefficient of $\boldsymbol{z}^a = z_0^{a_0} \cdots z_n^{a_n}$ in F. For $m \ge 1$, let $U_m := \{\zeta \in \mathbb{C} : \zeta^m = 1\}$, and let $\mathbf{1}_A$ be the indicator of a condition A.

The minimizing schedule of this section is in general complex, so we fix the meaning of a complex direction once. For $\boldsymbol{v} \in V_{\mathbb{C}}$ we define $T_p(\boldsymbol{x}; \boldsymbol{v})$ by the complex multilinear extension of $D^p u(\boldsymbol{x})$ on the right-hand side of (6). When u admits a holomorphic extension to a neighborhood of $\boldsymbol{x}$ in $\mathbb{C}^d$, this definition agrees with the order-p coefficient of $\tau \mapsto u(\boldsymbol{x} + \tau \boldsymbol{v})$, and the recurrences (8)–(10) evaluate it on complex tensors. When the directions are complex, the extension is guaranteed by entire activations such as $\sinh(z) = (e^z - e^{-z})/2$; see Remark 3.4. The distinction matters because an arbitrary C^p function on $\mathbb{R}^d$ need not be defined at complex arguments.

3.1. Basic expansion. Let $\boldsymbol{z} \in \mathbb{C}^{n+1}$ parametrize the active subspace through $\boldsymbol{v}(\boldsymbol{z}) := \sum_{j=0}^{n} z_j e_{i_j}$, and define the *generating polynomial* of u at $\boldsymbol{x}$ by

$$Q_u(\boldsymbol{z}) := T_p\big(\boldsymbol{x}; \boldsymbol{v}(\boldsymbol{z})\big). \tag{12}$$

The symmetric p-linearity of $D^p u(\boldsymbol{x})$ and the multinomial theorem give

$$\begin{aligned} D^p u(\boldsymbol{x})\big[\boldsymbol{v}(\boldsymbol{z}), \ldots, \boldsymbol{v}(\boldsymbol{z})\big] &= \sum_{|\beta|=p} \binom{p}{\beta} \boldsymbol{z}^\beta \, \partial^\beta u(\boldsymbol{x}) \\ &= \sum_{|\beta|=p} \frac{p!}{\beta!} \, \boldsymbol{z}^\beta \, \partial^\beta u(\boldsymbol{x}), \end{aligned}$$

where β ranges over multi-indices on the active variables and $\partial^\beta := \partial_{i_0}^{\beta_0} \cdots \partial_{i_n}^{\beta_n}$. Dividing by $p!$ and comparing with (6) and (12) yields

$$Q_u(\boldsymbol{z}) = \sum_{|\beta|=p} \frac{\partial^\beta u(\boldsymbol{x})}{\beta!} \, \boldsymbol{z}^\beta, \tag{13}$$

so Q_u is homogeneous of degree p and its coefficients are the order-p derivatives of u at $\boldsymbol{x}$. Reading off the coefficient of $\boldsymbol{z}^a$ in (13) and using $a! = \nu!$ from (11) gives

$$\partial^\nu u(\boldsymbol{x}) = \nu!\,[\boldsymbol{z}^a]\,Q_u(\boldsymbol{z}). \tag{14}$$

A single order-p derivative is therefore one coefficient of a homogeneous polynomial of degree p.

The evaluations available to us are values of Q_u at individual directions, so we ask for the shortest linear combination of such values that realizes the functional (14). A *directional schedule* of length R for ∂^ν is a finite collection $(c_r, \boldsymbol{v}_r)_{r=1}^R \subset \mathbb{C} \times V_{\mathbb{C}}$ such that

$$\partial^\nu u(\boldsymbol{x}) = \sum_{r=1}^{R} c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r) \tag{15}$$

for every u that is C^p on a neighborhood of $\boldsymbol{x}$. Because (15) is required for all such u, the order-p derivatives $(\partial^\beta u(\boldsymbol{x}))_{|\beta|=p}$ may be prescribed independently: for any coefficients q_β, the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(\boldsymbol{x}) = \beta!\, q_\beta$. Substituting (13) into the right-hand side of (15) gives

$$\sum_{r=1}^{R} c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(\boldsymbol{x})}{\beta!} \Big(\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta \Big), \tag{16}$$

and comparing (16) with the target $\partial^\nu u(\boldsymbol{x}) = \partial^a u(\boldsymbol{x})$ coefficient by coefficient shows that (15) holds for every admissible u if and only if

$$\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta = \nu!\, \delta_{a\beta}, \quad |\beta| = p, \tag{17}$$

where $\delta_{a\beta} := 1$ if $\beta = a$ and $\delta_{a\beta} := 0$ otherwise. The design problem is thus to solve the moment conditions (17) with as few directions as possible.

Two features of (17) decide how short a schedule can be. The conditions number $\binom{p+n}{n}$, so their count is governed by how many coordinates the derivative touches and not by the order alone: a derivative of order six on three coordinates imposes far fewer conditions than one of order six on six coordinates. They are also nonlinear in the unknowns, since each direction enters through the monomials $\boldsymbol{v}_r^\beta$, which is why the minimum is not read off by counting equations and unknowns. Section 3.2 resolves both by recognizing (17) as a polynomial identity whose shortest solution is classical.

3.2. Minimal schedules and monomial Waring rank. Pair a direction $\boldsymbol{v} \in V_{\mathbb{C}}$, written $\boldsymbol{v} = \sum_{j=0}^{n} v_j e_{i_j}$ in active coordinates, with the polynomial variable $\boldsymbol{z}$ through $\boldsymbol{v} \cdot \boldsymbol{z} := \sum_{j=0}^{n} v_j z_j$. Expanding the pth power of this pairing by the multinomial theorem gives

$$\sum_{r=1}^{R} c_r (\boldsymbol{v}_r \cdot \boldsymbol{z})^p = \sum_{|\beta|=p} \binom{p}{\beta} \Big(\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta \Big) \boldsymbol{z}^\beta, \tag{18}$$

whose coefficients are the very quantities constrained by (17). The moment conditions are therefore a polynomial identity in disguise, and the shortest schedule is the shortest decomposition of a monomial into pure powers of linear forms. That notion has a classical name.

Definition 3.1 (Waring rank). *Let $F \in \mathbb{C}[\boldsymbol{z}]_p$ be homogeneous of degree p. The* Waring rank *$R_{\mathbb{C}}(F)$ is the least R for which there exist $c_r \in \mathbb{C}$ and linear forms $L_r \in \mathbb{C}[\boldsymbol{z}]_1$ with $F = \sum_{r=1}^{R} c_r L_r(\boldsymbol{z})^p$.*

The following theorem is the main result of the paper. Its three parts identify the design problem with a Waring decomposition, evaluate the resulting minimum, and exhibit a schedule of that length in closed form.

Theorem 3.2 (Rank-optimal directional schedule). *Let ν be a multi-index of order $p \geq 1$ with active exponents $1 \leq a_0 \leq \cdots \leq a_n$ and $m_j := a_j + 1$ as in (11).*

(i) *A collection $(c_r, \boldsymbol{v}_r)_{r=1}^R \subset \mathbb{C} \times V_{\mathbb{C}}$ is a directional schedule for ∂^ν if and only if*

$$p!\, \boldsymbol{z}^a = \sum_{r=1}^{R} c_r (\boldsymbol{v}_r \cdot \boldsymbol{z})^p \quad \textit{in } \mathbb{C}[\boldsymbol{z}]_p. \tag{19}$$

(ii) *The minimum length of a directional schedule for ∂^ν equals $R_{\mathbb{C}}(\boldsymbol{z}^a)$, and this minimum is*

$$R_{\mathbb{C}}(\boldsymbol{z}^a) = \prod_{j=1}^{n} m_j = \frac{\prod_{j=0}^{n} m_j}{m_0}. \tag{20}$$

(iii) *For $\zeta = (\zeta_1, \ldots, \zeta_n) \in U_{m_1} \times \cdots \times U_{m_n}$, put*

$$\boldsymbol{v}_\zeta := e_{i_0} + \sum_{j=1}^{n} \zeta_j e_{i_j}, \quad c_\zeta := \frac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j. \tag{21}$$

Then $(c_\zeta, \boldsymbol{v}_\zeta)_\zeta$ is a directional schedule for ∂^ν of length $\prod_{j=1}^{n} m_j$, and by part (ii) no complex schedule is shorter.

Proof.

(i) The moment conditions are a polynomial identity. By (17), the collection is a directional schedule if and only if $\sum_r c_r \boldsymbol{v}_r^\beta = \nu!\, \delta_{a\beta}$ for every β with $|\beta| = p$. By (18), the identity (19) holds if and only if $\binom{p}{\beta} \sum_r c_r \boldsymbol{v}_r^\beta = p!\, \delta_{a\beta}$ for every such β. Dividing the second family of conditions by $\binom{p}{\beta}$ turns it into $\sum_r c_r \boldsymbol{v}_r^\beta = \beta!\, \delta_{a\beta}$, and $\beta!\, \delta_{a\beta} = \nu!\, \delta_{a\beta}$ by (11). The two families coincide, which proves part (i).

(ii) The minimum length is a Waring rank. Absorbing the nonzero constant $p!$ into the coefficients c_r shows that a decomposition of $p!\, \boldsymbol{z}^a$ into R pure pth powers exists if and only if one of $\boldsymbol{z}^a$ does, so $R_{\mathbb{C}}(p!\, \boldsymbol{z}^a) = R_{\mathbb{C}}(\boldsymbol{z}^a)$. Combining this with part (i) identifies the minimum schedule length with $R_{\mathbb{C}}(\boldsymbol{z}^a)$. The value (20) of the complex rank of a monomial is the theorem of Carlini, Catalisano, and Geramita [3]; Appendix A.3 recalls the apolarity argument behind its lower bound.

(iii) The roots-of-unity schedule attains the minimum. We use the conventions that an empty product equals 1 and that a Cartesian product over an empty index set has one element, the empty tuple; these cover the case $n = 0$, in which (21) produces the single direction e_{i_0} with coefficient $\nu!$. By part (i) it suffices to prove the identity $\sum_\zeta c_\zeta (\boldsymbol{v}_\zeta \cdot \boldsymbol{z})^p = p!\, \boldsymbol{z}^a$. Fix β with $|\beta| = p$ and let C_β denote the coefficient of

$\boldsymbol{z}^{\beta}$ on the left-hand side. Inserting (21) and applying the multinomial theorem gives

$$\begin{aligned} C_\beta &= \frac{\nu!}{\prod_{j=1}^n m_j}\binom{p}{\beta}\sum_{\zeta_1\in U_{m_1}}\cdots\sum_{\zeta_n\in U_{m_n}}\prod_{j=1}^n \zeta_j^{\beta_j+1} \\ &= \frac{\nu!}{\prod_{j=1}^n m_j}\binom{p}{\beta}\prod_{j=1}^n\Big(\sum_{\zeta_j\in U_{m_j}}\zeta_j^{\beta_j+1}\Big), \end{aligned} \tag{22}$$

where the second equality separates the independent sums. By Lemma A.1, the jth factor of the product in (22) vanishes unless $m_j \mid \beta_j + 1$, and $m_j = a_j + 1$ turns this divisibility into the congruence $\beta_j \equiv a_j \pmod{m_j}$. Suppose $C_\beta \neq 0$, and write $\beta_j = a_j + t_j m_j$ with $t_j \in \mathbb{N}_0$ for $j = 1, \ldots, n$. The degree constraint $|\beta| = p = \sum_{j=0}^n a_j$ then forces $\beta_0 = a_0 - \sum_{j=1}^n t_j m_j$. If some t_j were positive, then $m_j = a_j + 1 \geq a_0 + 1$ by the ordering of the exponents, whence $\beta_0 \leq a_0 - m_j < 0$, contradicting $\beta_0 \geq 0$. Every t_j therefore vanishes, so $\beta_j = a_j$ for $j \geq 1$, and the degree constraint gives $\beta_0 = a_0$ as well. Hence $C_\beta = 0$ for every $\beta \neq a$. For $\beta = a$ each inner sum in (22) equals m_j, the product cancels the prefactor, and

$$C_a = \nu!\binom{p}{a} = \Big(\prod_{j=0}^n a_j!\Big)\frac{p!}{\prod_{j=0}^n a_j!} = p!\,.$$

The left-hand side of the identity therefore equals $p!\,\boldsymbol{z}^a$, and the schedule has $|U_{m_1} \times \cdots \times U_{m_n}| = \prod_{j=1}^n m_j$ directions. ∎

Theorem 3.2 converts a question about derivative evaluation into a rank computation whose answer is the product (20). The minimizing directions and coefficients in (21) are roots of unity scaled by $\nu!/R_{\mathbb{C}}(\boldsymbol{z}^a)$, so their magnitudes are known in advance. Three consequences deserve separate mention.

Remark 3.3 (The smallest instance). *For $\partial^\nu = \partial_1\partial_2$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, and $U_2 = \{+1, -1\}$, so the roots of unity are real and (21) returns the two pairs $(\frac{1}{2}, e_1 + e_2)$ and $(-\frac{1}{2}, e_1 - e_2)$, that is, the elementary identity*

$$\partial_1\partial_2 u(\boldsymbol{x}) = \tfrac{1}{2}T_p(\boldsymbol{x}; e_1 + e_2) - \tfrac{1}{2}T_p(\boldsymbol{x}; e_1 - e_2), \tag{23}$$

in which the pure second derivatives cancel between the two probes. Theorem 3.2 adds to (23) the information that two evaluations are necessary as well as sufficient. Complex directions appear once two or more coordinates are active and some exponent is at least two. A pure power $\partial_1^p u$ uses one real direction, and a square-free monomial uses only the real pair $U_2 = \{\pm 1\}$.

Remark 3.4 (Entire activations are essential). *The directions (21) are complex whenever some factor U_{m_j} with $j \geq 1$ has $m_j \geq 3$, so the function is then evaluated at complex arguments and its activations must admit a holomorphic extension there. The functions* sinh *and* cosh *are entire, and the recurrence (10) is an algebraic identity that holds verbatim on complex tensors. By contrast* tanh *is meromorphic, with poles at $\mathrm{i}\pi/2 + \mathrm{i}k\pi$ for $k \in \mathbb{Z}$, so its jet is defined only away from those points and the admissible directions depend on the activation input. The collocation measurements of Section 4 use* sinh *at every order, including the order-two residuals whose directions remain real.*

Remark 3.5 (Scope of the rank reduction). *Theorem 3.2 concerns one monomial derivative at a time. For the square-free multi-index $a = (1, \ldots, 1)$ of order p the* *rank* (20) *equals* 2^{p-1}*, so the schedule is exponentially long; since the theorem gives a minimum, this is a statement about the problem and not about the construction, and no schedule of the form* (15) *can be shorter. A differential operator that is a linear combination of monomial derivatives is evaluated by applying the schedule to each monomial. The stochastic Taylor derivative estimator [26] treats the strictly broader problem of an arbitrary contraction of the order-*p *derivative tensor, with unbiased estimates whose variance is controlled by a sampling budget; the two approaches are complementary, and the schedule of Theorem 3.2 is the deterministic minimum for one fixed multi-index.*

Table 3.1 evaluates (20) for selected active-exponent patterns up to $p = 6$. Because every entry is a minimum, the table reads as a statement about which derivatives are cheap to extract from directional evaluations and which are not: a pure power needs one direction at any order, a derivative that repeats every index stays modest as the order grows, and a square-free derivative is intrinsically exponential. The explicit schedules for the patterns that occur in Section 4 are listed in Appendix A.2.

Table 3.1
The minimum number of directional evaluations $R_{\mathbb{C}}(\boldsymbol{z}^a)$ *of* (20)*, for selected active-exponent patterns up to* $p = 6$*. Repeating an index keeps the count low; the square-free pattern* (1^p) *is exponential in the order.*

pattern	example multi-index	$R_{\mathbb{C}}$
(p)	$\partial_1^p u$	1
$(2,1)$	u_{112}	3
$(1,1,1)$	u_{123}	4
$(3,1)$	u_{1112}	4
$(2,2)$	u_{1122}	3
$(2,1,1)$	u_{1123}	6
$(1,1,1,1)$	u_{1234}	8
$(4,2)$	u_{111122}	5
$(2,2,2)$	u_{112233}	9
(1^6)	u_{123456}	32

3.3. Algorithm, gradients, and cost. Algorithm 1 assembles the pieces. The schedule (21) is built once, its directions are stacked into a single batch, and one jet pass through the function returns the order-p coefficient for every direction at once; the reported value is then the coefficient-weighted sum. The schedule tensors depend on ν, the device, and the scalar type, but not on θ. The reference implementation builds them at call entry and includes that work in the measured throughput of Section 4, while a caller that reuses one multi-index may cache them. For a real-valued function the conjugate pairing of the roots of unity makes the returned value real in exact arithmetic; the implementation keeps the complex tensor and takes its real part only when the target PDE is real-valued.

Algorithm 1 must be differentiable in θ, since its output enters the residual term of (5). Differentiating the activation recurrences (9)–(10) by reverse-mode automatic differentiation would rebuild inside the backward pass the nesting that the jet was introduced to avoid. We therefore register each activation jet as a single primitive with the closed-form adjoint

$$g_x^{(m)} = \sum_{j=0}^{p-m} \overline{q_j}\, g_y^{(m+j)}, \quad q(\tau) := \phi'\big(x(\tau)\big), \tag{24}$$

Algorithm 1 Single-monomial $\partial^\nu u(\boldsymbol{x})$ from a rank-optimal schedule and one batched Taylor jet.

Require: scalar network u built from supported analytic primitives; point $\boldsymbol{x} \in \mathbb{R}^d$; multi-index ν of order p
Ensure: the value $\partial^\nu u(\boldsymbol{x})$

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Relabel the active coordinates of ν as i_0, ..., i_n with exponents 1 ≤ a_0 ≤ ··· ≤ a_n.
Set m_j := a_j + 1 for j = 1, ..., n and I := U_{m_1} × ··· × U_{m_n}.
Initialize S ← ∅.
for ζ = (ζ_1, ..., ζ_n) ∈ I do
    Set v_ζ ← e_{i_0} + Σ_{j=1}^n ζ_j e_{i_j} and c_ζ ← ν! (Π_{j=1}^n m_j)^{-1} Π_{j=1}^n ζ_j.
    Update S ← S ∪ {(c_ζ, v_ζ)}.
end for
Stack the directions of S into V_S ∈ C^{|S|×d}.
Form the batched input jet J_0 ← (x replicated, V_S, 0, ..., 0).
Propagate J_0 through u by (8)–(10) to obtain J_L.
Set t ← [J_L]_p, one order-p coefficient per direction.
return Σ_{r=1}^{|S|} c_r t_r.
```

in which q_j is the jth Taylor coefficient of q and $g^{(m)}$ denotes the cotangent of the order-m coefficient. Equation (24) follows from the chain rule applied to the jet of ϕ and uses the convention of a real-valued loss for complex parameters, so that the conjugation is the Hermitian adjoint of multiplication by q_j. The backward pass thus costs one more triangular convolution per activation and never differentiates through the recurrence itself.

The cost of Algorithm 1 separates into a per-direction and a per-schedule factor. For one direction, the jet stores $p+1$ coefficient tensors per activation, an affine layer applies W to each of them, and the activation recurrences require $\sum_{k=1}^{p} O(k) = O(p^2)$ elementwise products per scalar. The total work and the directional storage then scale linearly in the schedule length $R_{\mathbb{C}}(\boldsymbol{z}^a)$, which by Theorem 3.2 is as small as an exact directional formula can be. The rank reduction is an exact statement about that factor and not about wall-clock time, which also depends on complex arithmetic, kernel launch overhead, memory traffic, and whether parameter gradients are required; for small widths or small batches these terms can dominate. Section 4 therefore reports measured throughput alongside accuracy.

The schedule coefficients are known in closed form. The remaining error is floating-point round-off from the Taylor recurrences, from complex arithmetic, and from the final summation over directions. Section 4.1 reports double-precision agreement through order six.

4. Numerical experiments. This section checks two properties of the schedule: agreement with nested automatic differentiation in double precision, and the time and memory of one residual evaluation, including the reverse-mode pass. The first comparison uses no PDE. The second repeats the same evaluation inside an otherwise identical collocation loop and records the time and memory of one step. Every quantity below is a property of the derivative method on a fixed function, residual, and optimizer.

4.1. Verification of the derivative backend. All checks use double precision and are executable tests in the repository. The roots-of-unity coefficient filter of Lemma A.1 is evaluated against every degree-four monomial in five variables for the active-exponent pattern $(2, 1, 1)$, with a maximum admissible absolute coefficient error of 10^{-12}. The length of the schedule that the implementation builds is checked against the closed-form rank (20) for every pattern of Table 3.1.

The Waring jet of Algorithm 1 is then compared with nested automatic differentiation on real-parameter networks using the tanh, logistic, sin, and sinh activations, for the patterns $(2, 2)$, $(1, 1, 1)$, and (3). Values must agree to relative tolerance 2×10^{-10} and absolute tolerance 2×10^{-11}, and the imaginary remainder of a real target must not exceed 2×10^{-11}. Parameter gradients of a squared derivative loss are compared at tolerances 2×10^{-9} relative and 2×10^{-11} absolute, on a real tanh network and on a complex sinh network; additional analytic checks cover the orders 1, 2, 3, 4, and 6 for a scalar complex sinh network. Every check passes at the tolerance stated for it. These tests establish that the implementation computes what Theorem 3.2 predicts.

4.2. A controlled comparison of derivative backends. The remaining study isolates the derivative backend. Protocol `jsc_v3` runs two lines that share the network, the loss, the optimizer, the collocation points, the evaluation rule, the seeds, and the wall-clock budget, and differ in one respect: how each monomial derivative of the residual is evaluated. The first line uses Algorithm 1, that is, the rank-optimal complex schedule evaluated by one batched Taylor jet, and is labeled *jet* below. The second evaluates the same derivative of the same network by nested reverse-mode automatic differentiation, with the graph retained after every coordinate derivative, and is labeled *nested*. Both lines are the same complex-parameter sinh network with four hidden layers of width $H = 128$, hence the same 100,098 trainable real degrees of freedom, the same first-layer scale, and the same `complex128` arithmetic. Both use Adam with initial learning rate 10^{-3} and cosine decay, 4096 fixed interior points, 512 fixed boundary points, paired per-seed collocation, the same per-setting boundary weights, and a common held-out set of 8192 points. Each line is run for three seeds with a budget of 1000 seconds per run.

A *setting* is one problem together with one value of its sweep parameter, and the protocol covers nine: the two-dimensional polyharmonic problem at orders $2m \in \{2, 4, 6\}$, the radial chirp at $a \in \{1, 2, 3\}$, and the time-harmonic Maxwell problem at $a \in \{2, 4, 6\}$, all defined as in Section 4.3. All 54 runs were produced by one source commit on one NVIDIA H20 GPU with PyTorch 2.5.1 and CUDA 12.1, so unlike a comparison assembled from separate campaigns, timings here are comparable across settings as well as within one. Cost is the measured time per optimizer step together with the peak memory allocated by a run, and accuracy is the held-out relative L^2 error

$$\mathcal{E}_{L^2} := \frac{\|u_\theta - u_\star\|_{L^2(\Omega)}}{\|u_\star\|_{L^2(\Omega)}}, \tag{25}$$

evaluated on that held-out set against the exact solution $u_\star$. A setting enters the tables only after automated checks confirm all six method-seed rows, finite metrics, nonempty monotone histories, and the protocol metadata.

4.3. Benchmark problems. The three families place different demands on the derivative. For the polyharmonic problem [5, 7] on $(-1, 1)^2$ we solve

$$\Delta^m u = (-2\pi^2)^m u_\star \quad \text{in } (-1, 1)^2, \quad u_\star(\boldsymbol{x}) = \sin(\pi x_1)\sin(\pi x_2), \tag{26}$$

at the orders $2m \in \{2,4,6\}$, with the Navier boundary targets $u = u_\star$ and $\Delta^j u = (-2\pi^2)^j u_\star$ for $j = 1,\dots,m-1$. Those traces vanish on $\partial(-1,1)^2$ because $u_\star$ does. This family is the reason the derivative order appears as a variable: expanding Δ^m in two dimensions produces the patterns $(2m)$ and, for $m \ge 2$, the mixed patterns whose schedules Table 3.1 lists. At $2m = 4$ the mixed term is $\partial_1^2\partial_2^2$, which (21) evaluates with three directions; at $2m = 6$ the mixed terms are $\partial_1^4\partial_2^2$ and $\partial_1^2\partial_2^4$, which take five each.

The other two families sit at order two and sweep a frequency instead. On $(-1,1)^2$ the radial chirp [16] is

$$-\Delta u + u = f, \quad u_\star(x,y) = \sin\Big(\frac{a\pi}{2}(x^2+y^2)\Big), \tag{27}$$

with Dirichlet data from $u_\star$, $a \in \{1,2,3\}$, and the manufactured source $f = (a\pi)^2 r^2 \sin\phi - 2a\pi\cos\phi + \sin\phi$, where $\phi := a\pi(x^2+y^2)/2$ and $r^2 := x^2+y^2$. Its local radial wavenumber $|\nabla\phi| = a\pi r$ varies over the domain, so the target is not a separable Fourier mode. For a two-dimensional transverse-magnetic mode, Maxwell's equations [21] reduce to a scalar complex Helmholtz equation for the out-of-plane field E,

$$\Delta E + \kappa^2 E = f, \quad \kappa^2 = (a\pi)^2(1+\mathrm{i}\beta), \quad E_\star = \exp\big(\mathrm{i}a\pi(x+y)\big), \tag{28}$$

with $\beta = 0.2$, exact Dirichlet data, $a \in \{2,4,6\}$, and $f = [-2(a\pi)^2 + \kappa^2]E_\star$. The Maxwell family has a complex-valued solution. The measurements already propagate complex tensors through a complex-parameter sinh network, so a complex residual uses the same jet as the real ones. Since both families are of order two, every monomial term of their residuals is a pure power, which the schedule evaluates with a single real direction. They therefore isolate the cost of the Taylor jet on one direction.

Table 4.1

Cost of one optimizer step under the jsc_v3 protocol, which changes only the derivative backend. Times are the three-seed mean of milliseconds per step and memory is the peak allocation of a run; the ratio columns give nested automatic differentiation divided by the Taylor jet, so larger is better for the jet.

		ms per step			peak memory (MiB)		
setting	p	jet	nested	ratio	jet	nested	ratio
Polyharmonic, $2m=2$	2	62.3	91.1	**1.46**	509	789	**1.55**
Polyharmonic, $2m=4$	4	251.6	565.9	**2.25**	1759	5400	**3.07**
Polyharmonic, $2m=6$	6	854.2	3496.1	**4.09**	5728	36976	**6.46**
Radial chirp, $a=1$	2	72.8	101.7	**1.40**	548	852	**1.55**
Radial chirp, $a=2$	2	73.4	101.9	**1.39**	548	852	**1.55**
Radial chirp, $a=3$	2	73.4	102.1	**1.39**	548	852	**1.55**
Maxwell, $a=2$	2	73.3	102.1	**1.39**	573	853	**1.49**
Maxwell, $a=4$	2	73.2	101.9	**1.39**	573	853	**1.49**
Maxwell, $a=6$	2	72.9	101.8	**1.40**	573	853	**1.49**

4.4. Cost of one derivative evaluation. Table 4.1 and Figure 4.1 report the cost of one optimizer step. At order two the jet is faster by a factor between 1.39 and 1.46 across all seven order-two settings, and it allocates between 1.49 and 1.55 times less memory. At order two every residual term is a pure power, so the schedule uses one direction and the saving is that of Taylor-mode evaluation: nested differentiation retains a graph across two coordinate derivatives. The saving then grows with the

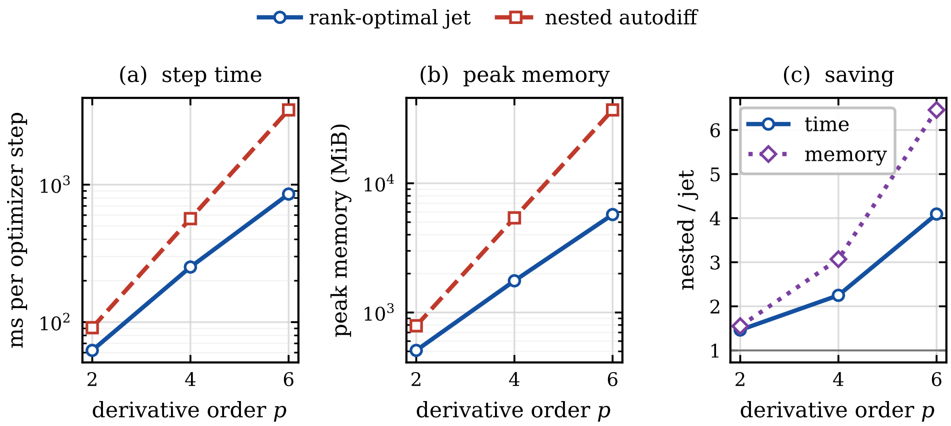

FIG. 4.1. *Cost of one optimizer step against the derivative order, on the polyharmonic family, for the two backends of Section 4.2. Panels (a) and (b) show the three-seed mean step time and the peak memory of a run; panel (c) shows the ratio of nested automatic differentiation to the Taylor jet for both quantities.*

order, exactly where the rank reduction begins to apply: at order four the step time falls by a factor of 2.25 and the peak memory by 3.07, and at order six by 4.09 and 6.46. In absolute terms, one step of the sixth-order problem costs 854 milliseconds and 5.6 gibibytes with the jet, against 3.5 seconds and 36.1 gibibytes with nested differentiation.

Two mechanisms compound in that growth, and the design of the study separates them. The jet replaces a graph of depth proportional to the derivative order and the network depth by $p+1$ coefficient tensors per activation, which is what the memory column measures; this part is available at every order and is what the order-two settings show. The schedule then fixes how many times that jet must be run, at the minimum of Table 3.1: one direction for each pure power at any order, three for the mixed term at order four, five for each mixed term at order six. Nested differentiation has no comparable lever, since it must traverse the retained graph once per coordinate derivative of every term, which is why the ratio in panel (c) of Figure 4.1 widens with the order. Both counts are exact. Table 4.1 shows that on this hardware the reduction survives complex arithmetic, kernel launches, and the backward pass.

4.5. What the saving buys. A cheaper evaluation is useful if more evaluations fit in the same wall-clock budget. Table 4.2 therefore records how many optimizer steps each line completes inside the common budget and the accuracy it reaches, and Figure 4.2 shows the trajectories. The step counts follow the cost directly: at order six the jet completes 1171 steps against 287, and at order four 3975 against 1767. The accuracy follows the step counts where the problem is hard enough for the extra steps to matter. At order four the jet reaches 6.55×10^{-4} against 1.43×10^{-3}, and at order six 3.36×10^{-2} against 9.18×10^{-2}, in both cases with the seed spread of Table 4.2 well inside the gap. Across the nine settings the jet has the lower three-seed mean error in eight; the exception is the chirp at $a = 2$, where the two lines are within one seed standard deviation of each other.

At order two the extra steps buy little: the two backends track each other in panels (b) and (c) of Figure 4.2, and in five of the six frequency-swept settings the

TABLE 4.2
What the saved time buys under the same protocol: optimizer steps completed inside the common 1000-second budget, as a three-seed mean, and the held-out relative L^2 error reached, as a three-seed mean $\pm$ sample standard deviation. The lower error in each row is set in bold.

setting	steps completed jet	steps completed nested	relative L^2 error jet	relative L^2 error nested
Polyharmonic, $2m = 2$	16058	10973	$\mathbf{3.25 \pm 1.15{\times}10^{-4}}$	$5.42 \pm 0.88{\times}10^{-4}$
Polyharmonic, $2m = 4$	3975	1767	$\mathbf{6.55 \pm 0.71{\times}10^{-4}}$	$1.43 \pm 0.10{\times}10^{-3}$
Polyharmonic, $2m = 6$	1171	287	$\mathbf{3.36 \pm 1.81{\times}10^{-2}}$	$9.18 \pm 3.62{\times}10^{-2}$
Radial chirp, $a = 1$	13730	9835	$\mathbf{3.84 \pm 3.68{\times}10^{-4}}$	$7.05 \pm 3.55{\times}10^{-4}$
Radial chirp, $a = 2$	13617	9816	$8.93 \pm 10.97{\times}10^{-4}$	$\mathbf{8.36 \pm 4.56{\times}10^{-4}}$
Radial chirp, $a = 3$	13629	9796	$\mathbf{1.86 \pm 0.23{\times}10^{-4}}$	$3.38 \pm 1.25{\times}10^{-4}$
Maxwell, $a = 2$	13639	9795	$\mathbf{1.24 \pm 0.44{\times}10^{-3}}$	$1.61 \pm 0.72{\times}10^{-3}$
Maxwell, $a = 4$	13665	9814	$\mathbf{1.24 \pm 0.40{\times}10^{-3}}$	$1.48 \pm 0.67{\times}10^{-3}$
Maxwell, $a = 6$	13712	9825	$\mathbf{9.27 \pm 1.52{\times}10^{-3}}$	$9.42 \pm 1.49{\times}10^{-3}$

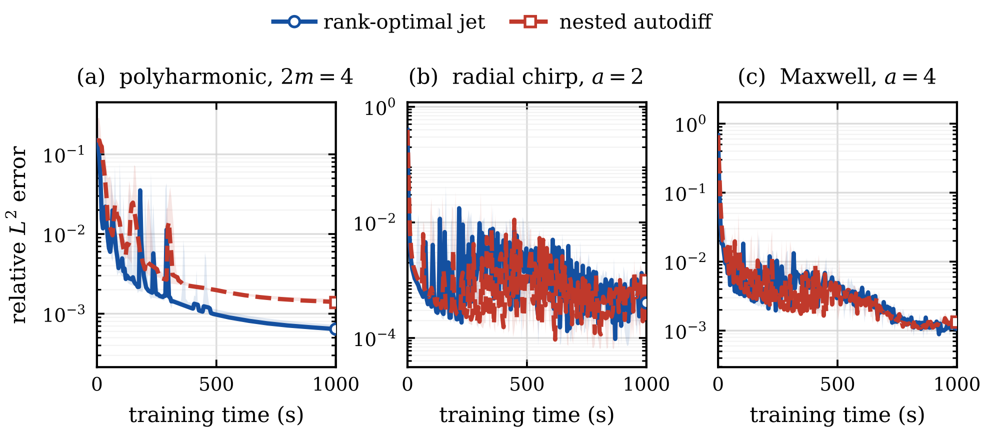

FIG. 4.2. *Held-out relative L^2 error against training time for the two backends, on one representative setting of each family. Curves are the three-seed median and the shaded band is the seedwise range. Each panel is framed on the median curves.*

two means differ by less than one seed standard deviation. A cheaper derivative is not by itself a better optimizer, and at low order the derivative is not what limits the run. The cost advantage of the schedule grows with the derivative order, in wall time and in memory, with the model, the loss, and the optimizer held fixed.

5. Conclusion. The shortest exact formula for a high-order derivative, in directional Taylor coefficients of the same order, is the Waring decomposition of the corresponding monomial. Carlini, Catalisano, and Geramita evaluate the complex length as (20), and a roots-of-unity formula attains it. No shorter linear combination of order-p directional Taylor coefficients exists; the two-point identity for $\partial_1\partial_2$ is the length-two case.

Combined with a batched Taylor jet and a closed-form vector-Jacobian product, the schedule evaluates the derivative exactly and remains differentiable in θ. Double-precision tests reproduce nested automatic differentiation for values and for parameter gradients through order six. Measured against nested differentiation, with the function, the loss, and the optimizer held fixed, the schedule reduces the time of one evaluation by factors of about 1.4, 2.3, and 4.1 at derivative orders two, four, and six, and its peak memory by about 1.5, 3.1, and 6.5; on the higher-order problems the evaluations thus gained lower the error reached within a fixed budget. The two mechanisms behind that growth are separable in the measurements: Taylor-mode evaluation removes the nested graph at every order, and the rank reduction shortens the schedule wherever an index repeats.

The schedule is shortest for repeated-index multi-indices and exponentially long for square-free ones: Theorem 3.2 shows that this length is the minimum. At low derivative order a cheaper derivative also buys little in a complete run, since the derivative is not what limits it. Natural continuations are a rank-aware expansion of an operator that reuses directions across its terms, a conditioning analysis of the schedule at high order and in reduced precision, and the extension of the same apolarity argument to derivative tensors contracted against a fixed structured family.

Appendix A. Schedules, filters, and the apolar bound.

This appendix collects the roots-of-unity filter used in the proof of Theorem 3.2, the explicit schedules for the derivative patterns that occur in Section 4, and the apolarity argument behind the rank formula (20). The formulas of Appendix A.2 are those the implementation evaluates.

A.1. Roots-of-unity filter. Lemma A.1. *For every integer $m \geq 1$ and every integer k,*

$$\sum_{\zeta \in U_m} \zeta^k = m\,\mathbf{1}_{m|k}. \tag{29}$$

Proof. Let $\omega := \exp(2\pi \mathrm{i}/m)$, so that every element of U_m is ω^ℓ for exactly one $\ell \in \{0, \dots, m-1\}$ and

$$\sum_{\zeta \in U_m} \zeta^k = \sum_{\ell=0}^{m-1} \left(\omega^k\right)^\ell.$$

If $m \mid k$, then $\omega^k = 1$ and the sum equals m. If $m \nmid k$, then $\omega^k \neq 1$, and the finite geometric series gives

$$\sum_{\ell=0}^{m-1} \left(\omega^k\right)^\ell = \frac{1-\omega^{km}}{1-\omega^k} = 0,$$

since $\omega^m = 1$. ∎

A.2. Explicit low-order schedules. The mixed terms of the polyharmonic residual (26) and the pure powers of the chirp and Maxwell residuals use the following instances of (21).

For a pure power $\partial^\nu = \partial_1^p$ we have $n = 0$, the schedule has one direction, and T_p is evaluated once,

$$\partial_1^p u(\boldsymbol{x}) = p!\, T_p(\boldsymbol{x}; e_1). \tag{30}$$

For the fourth-order mixed term $\partial^\nu = \partial_1^2 \partial_2^2$ we have $n = 1$, $a = (2,2)$, and $m_1 = 3$; with $\omega := \exp(2\pi \mathrm{i}/3)$ and $U_3 = \{1, \omega, \omega^2\}$, formula (21) gives the three directions

$$\partial_1^2 \partial_2^2 u(\boldsymbol{x}) = \frac{4}{3} \sum_{\zeta \in U_3} \zeta\, T_p(\boldsymbol{x};\, e_1 + \zeta e_2). \tag{31}$$

For the sixth-order mixed term $\partial^\nu = \partial_1^2\partial_2^4$ we have $n = 1$, $a = (2, 4)$, and $m_1 = 5$; with U_5 the fifth roots of unity, (21) takes the coordinate of smaller exponent as base and gives the five directions

$$\partial_1^2\partial_2^4 u(\boldsymbol{x}) = \frac{48}{5} \sum_{\zeta\in U_5} \zeta\, T_p(\boldsymbol{x};\, e_1 + \zeta e_2). \tag{32}$$

The companion term $\partial_1^4\partial_2^2$ is the same formula with the two coordinates exchanged. The equivalent polynomial identity (19) for (32) reads

$$6!\, z_0^2 z_1^4 = \frac{48}{5} \sum_{\zeta\in U_5} \zeta\,(z_0 + \zeta z_1)^6, \tag{33}$$

and it can be checked directly: the coefficient of $z_0^{\beta_0} z_1^{\beta_1}$ on the right-hand side contains the factor $\sum_{\zeta\in U_5} \zeta^{1+\beta_1}$, which by Lemma A.1 is nonzero only when $\beta_1 \equiv 4 \pmod 5$. Together with $\beta_0 + \beta_1 = 6$ this forces $\beta = (2, 4)$, and the surviving constant is $\frac{48}{5} \cdot \frac{6!}{2!\,4!} \cdot 5 = 720 = 6!$. The three-variable pattern $(2, 2, 2)$ of Table 3.1 is the same construction with two cube-root factors and nine directions.

A.3. Apolar form of the rank bound. The lower bound in (20) comes from apolarity, which we recall for completeness. For a homogeneous $F \in \mathbb{C}[\boldsymbol{z}]_p$, let $F^\perp$ denote its apolar ideal, that is, the ideal of differential operators annihilating F, and call a finite subscheme $X \subset \mathbb{P}^n$ apolar to F when $I_X \subset F^\perp$. The Waring rank then admits the characterization

$$R_{\mathbb{C}}(F) = \min\bigl\{\deg X : X \subset \mathbb{P}^n \text{ finite, reduced, and apolar to } F\bigr\}. \tag{34}$$

For the monomial $F = \boldsymbol{z}^a$ the apolar ideal is the complete intersection

$$(\boldsymbol{z}^a)^\perp = \bigl(\partial_0^{a_0+1}, \dots, \partial_n^{a_n+1}\bigr), \tag{35}$$

and the Hilbert function of the quotient $\mathbb{C}[\partial]/(\boldsymbol{z}^a)^\perp$ bounds the degree of an apolar subscheme from below by the product $\prod_{j=1}^n m_j$ of (20); see [15, 19] for the general theory and [3] for the monomial case. The construction (21) exhibits a reduced apolar subscheme of exactly that degree, so the bound is attained.

REFERENCES

[1] Atilim Gunes Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul, and Jeffrey Mark Siskind, *Automatic differentiation in machine learning: a survey*, Journal of Machine Learning Research, 18 (2018), pp. 1–43.

[2] Jesse Bettencourt, Matthew J. Johnson, and David Duvenaud, *Taylor-mode automatic differentiation for higher-order derivatives in JAX*, in NeurIPS Workshop on Program Transformations for Machine Learning, 2019.

[3] Enrico Carlini, Maria Virginia Catalisano, and Anthony V. Geramita, *The solution to the Waring problem for monomials and the sum of coprime monomials*, Journal of Algebra, 370 (2012), pp. 5–14.

[4] Pao-Hsiung Chiu, Jian Cheng Wong, Chinchun Ooi, My Ha Dao, and Yew-Soon Ong, *CAN-PINN: A fast physics-informed neural network based on coupled-automatic–numerical differentiation method*, Computer Methods in Applied Mechanics and Engineering, 395 (2022), p. 114909.

[5] Philippe G. Ciarlet, *Mathematical Elasticity, Volume II: Theory of Plates*, vol. 27 of Studies in Mathematics and its Applications, North-Holland, Amsterdam, 1997.

[6] Bengt Fornberg, *Generation of finite difference formulas on arbitrarily spaced grids*, Mathematics of Computation, 51 (1988), pp. 699–706.

[7] Filippo Gazzola, Hans-Christoph Grunau, and Guido Sweers, *Polyharmonic Boundary Value Problems*, vol. 1991 of Lecture Notes in Mathematics, Springer, Berlin, 2010.
[8] Andreas Griewank, Jean Utke, and Andrea Walther, *Evaluating higher derivative tensors by forward propagation of univariate Taylor series*, Mathematics of Computation, 69 (2000), pp. 1117–1130.
[9] Andreas Griewank and Andrea Walther, *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2nd ed., 2008.
[10] Di He, Shanda Li, Wenlei Shi, Xiaotian Gao, Jia Zhang, Jiang Bian, Liwei Wang, and Tie-Yan Liu, *Learning physics-informed neural networks without stacked backpropagation*, in Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, vol. 206 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 3034–3047.
[11] Zheyuan Hu, Zekun Shi, George Em Karniadakis, and Kenji Kawaguchi, *Hutchinson trace estimation for high-dimensional and high-order physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering, 424 (2024), p. 116883.
[12] ———, *Bias-variance trade-off in physics-informed neural networks with randomized smoothing for high-dimensional PDEs*, Journal of Computational Physics, 521 (2025), p. 113524.
[13] Zheyuan Hu, Khemraj Shukla, George Em Karniadakis, and Kenji Kawaguchi, *Tackling the curse of dimensionality with physics-informed neural networks*, Neural Networks, 176 (2024), p. 106369.
[14] Michael F. Hutchinson, *A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines*, Communications in Statistics: Simulation and Computation, 18 (1989), pp. 1059–1076.
[15] Anthony Iarrobino and Vassil Kanev, *Power Sums, Gorenstein Algebras, and Determinantal Loci*, vol. 1721 of Lecture Notes in Mathematics, Springer, Berlin, 1999.
[16] Frank Ihlenburg, *Finite Element Analysis of Acoustic Scattering*, vol. 132 of Applied Mathematical Sciences, Springer, New York, 1998.
[17] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang, *Physics-informed machine learning*, Nature Reviews Physics, 3 (2021), pp. 422–440.
[18] Isaac E. Lagaris, Aristidis Likas, and Dimitrios I. Fotiadis, *Artificial neural networks for solving ordinary and partial differential equations*, IEEE Transactions on Neural Networks, 9 (1998), pp. 987–1000.
[19] Joseph M. Landsberg, *Tensors: Geometry and Applications*, vol. 128 of Graduate Studies in Mathematics, American Mathematical Society, Providence, RI, 2012.
[20] Ruichen Li, Haotian Ye, Du Jiang, Xuelan Wen, Chuwei Wang, Zhe Li, Xiang Li, Di He, Ji Chen, Weiluo Ren, and Liwei Wang, *A computational framework for neural network-based variational Monte Carlo with Forward Laplacian*, Nature Machine Intelligence, 6 (2024), pp. 209–219.
[21] Peter Monk, *Finite Element Methods for Maxwell's Equations*, Oxford University Press, Oxford, 2003.
[22] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala, *PyTorch: An imperative style, high-performance deep learning library*, in Advances in Neural Information Processing Systems, vol. 32, 2019.
[23] Barak A. Pearlmutter, *Fast exact multiplication by the Hessian*, Neural Computation, 6 (1994), pp. 147–160.
[24] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics, 378 (2019), pp. 686–707.
[25] Patrick J. Roache, *Code verification by the method of manufactured solutions*, Journal of Fluids Engineering, 124 (2002), pp. 4–10.
[26] Zekun Shi, Zheyuan Hu, Min Lin, and Kenji Kawaguchi, *Stochastic Taylor derivative estimator: Efficient amortization for arbitrary differential operators*, Advances in Neural Information Processing Systems (NeurIPS), 37 (2024), pp. 122316–122353.
[27] Justin Sirignano and Konstantinos Spiliopoulos, *DGM: A deep learning algorithm for solving partial differential equations*, Journal of Computational Physics, 375 (2018), pp. 1339–1364.

[28] Charles M. Stein, *Estimation of the mean of a multivariate normal distribution*, The Annals of Statistics, 9 (1981), pp. 1135–1151.